

Hybridization

It is a hypothetical concept. Orbitals of atom having similar energy intermix and redistribute their energy and form a new orbitals. These orbitals are called hybrid orbitals and this phenomenon is called hybridization. Hybrid orbitals may be vacant, half filled, full filled. Hybrid orbitals never form π bond but benzene is exception.



this π bond is formed by overlap of $sp^2 - sp^2$ orbital.

Types of hybridization.

Steric No: (CN)	hybridization	Geometry	orbitals Involved.
2	sp	linear	S, p_z or p_x or p_y
3	sp^2	Trigonal planar	S, p_x, p_y
4	sp^3	Tetrahedral	S, p_x, p_y, p_z

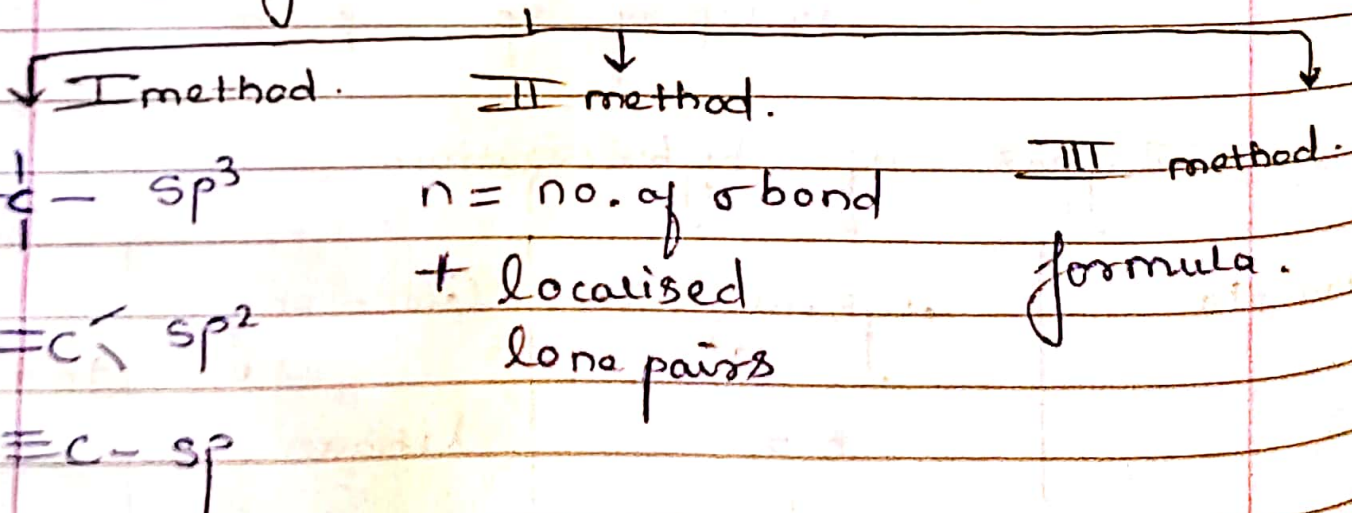
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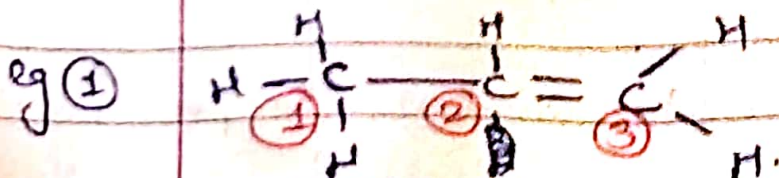
Steric No: (N)	Hyb ⁿ	Geometry	orbitals involved
4	dsp^2	Sq. planar	dx^2-y^2, s, p_x, p_y
5	sp^3d	Trigonal bipyramidal	$s, p_x, p_y, p_z, dz^2, dx^2-y^2$
6	sp^3d^2	Octahedral	$s, p_x, p_y, p_z, dx^2-y^2, dz^2$
7	sp^3d^3	Pentagonal bipyramidal	$s, p_x, p_y, p_z, dx^2-y^2, dz^2, dxz, dyz$

Identification



I method:-

Count σ bond.



(3)

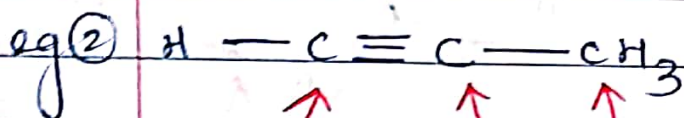
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(1) C has 4 σ bonds; $N=4$; $\therefore sp^3$

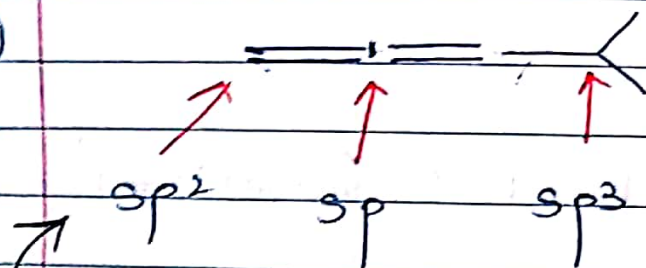
(2) C has 3 σ bond; $N=3$; $\therefore sp^2$
(don't count π bond, hybrid orbitals
now do not form π bonds)

(3) C has 3 σ bonds; $N=3$; $\therefore sp^2$

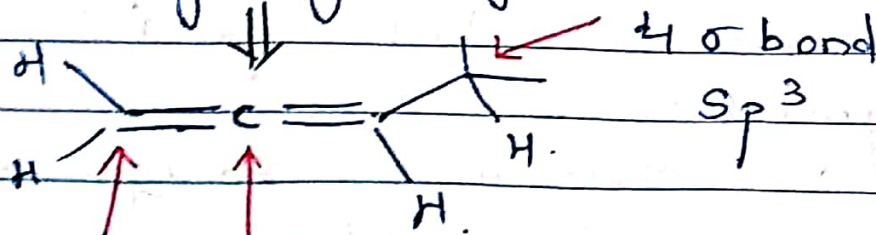


2 σ bond sp 2 σ bond $N=2$ sp 4 σ bond $N=4$ sp^3

eg(3)



for such type of bond line notations
first fulfill the valency with
hydrogens for each carbon.

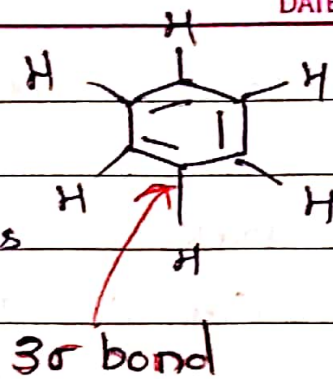


(4)

eg (4)



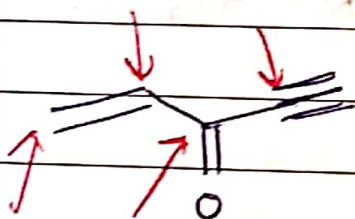
draw
Hydrogens



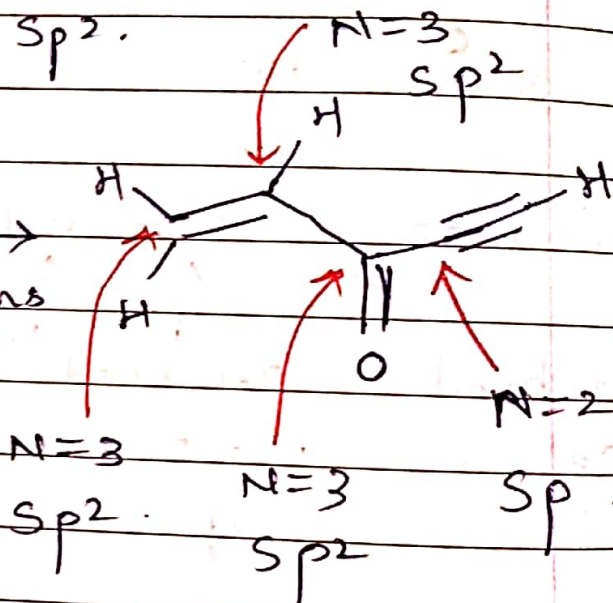
$n=3$

sp^2

eg (5)



draw
Hydrogens



III method

$N = \text{no. of } \sigma \text{ bond} + \text{localised lone pairs.}$

{ localised lone pairs are the lone pairs that remain at their place and do not take part in Conjugation.

while delocalized lone pairs are the part of Conjugation }

5



pyrrole.

what is the hybridization of N.

Step 1 :- first see whether all the valencies of N are complete, if not then complete it with H.



$N = \text{no. of } \sigma \text{ bonds} + \text{localized l.p.s.}$

Since this l.p. is

$$N = 3 + 0$$

$$N = 3 \Rightarrow sp^2$$

part of conjugation it is delocalised. and is not counted in formula.

eg 2)



$N = \text{no. of } \sigma \text{ bonds} + \text{localized l.p.}$

$$N = 2 + 1 ; N = 3 \Rightarrow sp^2$$

furane

{ out of two lone pairs in furane one is delocalized $\therefore$ it is not counted }

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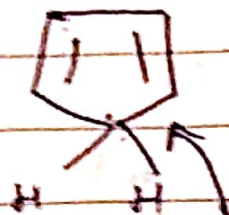
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eg 3)



draw

Hydrogens



$$N = 4 + 0; N = 4 \Rightarrow sp^3$$

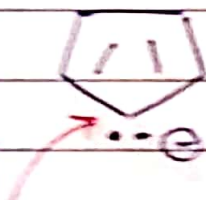
eg 4)



the lone pair is localized

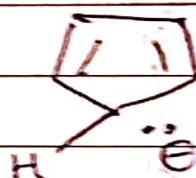
$$N = 2 + 1; N = 3 \Rightarrow sp^2$$

eg 5)



draw

Hydrogen



l.p is delocalized.

$$N = 3 + 0 \Rightarrow 3 \Rightarrow sp^2$$

Method III

Formula.

$N = \text{Number of valence shell electrons around central atom} + \text{no. of monovalent surrounding atoms} \pm \text{charge}$

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if molecule is having +ve charge then it is subtracted in formula and if there is -ve charge; it is added in formula.

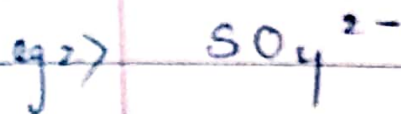


step 1

Identify central atom.
here it is P.

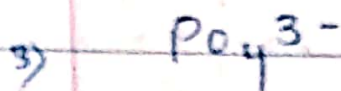
step

$$N = \frac{5 + 5 + 0}{2} \Rightarrow N = 5 \Rightarrow sp^3d$$



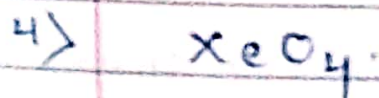
$$N = \frac{6 + 2}{2} \Rightarrow 4 \Rightarrow sp^3$$

charge



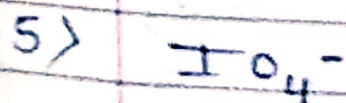
$$N = \frac{5 + 3}{2} \Rightarrow 4 \Rightarrow sp^3$$

charge

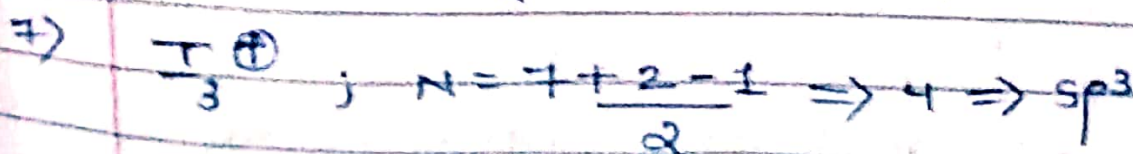
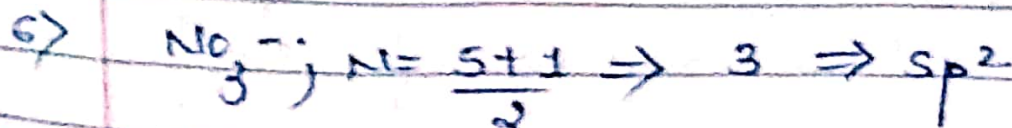


$$N = \frac{8}{2} \Rightarrow 4 \Rightarrow sp^3$$

{ Since oxygen is divalent; it is not counted in formula }



$$N = \frac{7 + 1}{2} \Rightarrow sp^3$$



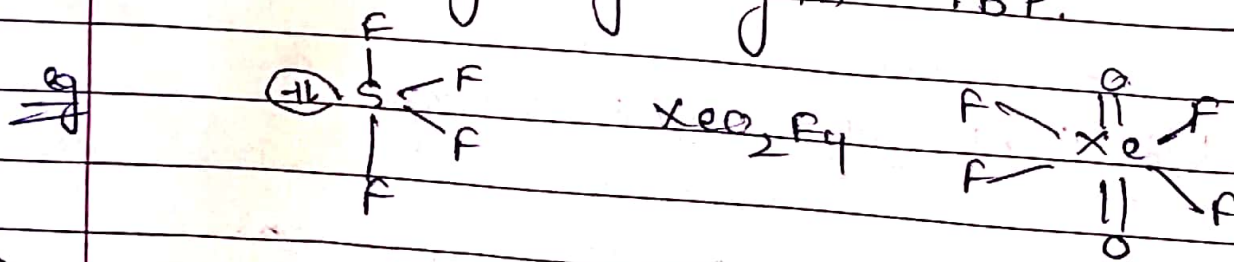
Valence Shell electron pair repulsion theory (VSEPR)

Valence Shell electron pair repulsion theory was given by Sidgwick and Powell and was further extended by Gillespie and Nyholm.

According to this theory valence shell e-pairs (l.p. as well as b.p.) play an important role in deciding shape or geometry of molecules.

Important points regarding VSEPR theory

- 1) $\text{l.p.} - \text{l.p.} > \text{l.p.} - \text{b.p.} > \text{b.p.} - \text{b.p.}$
repulsion repulsion repulsion.
- 2) lone pairs and multiple bonds tend to occupy more space, therefore they occupy positions having more bond angle. for eg in TBP.



- 3) Size and electronegativity of central as well as surrounding species play important

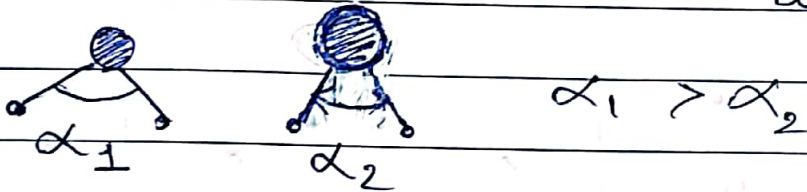
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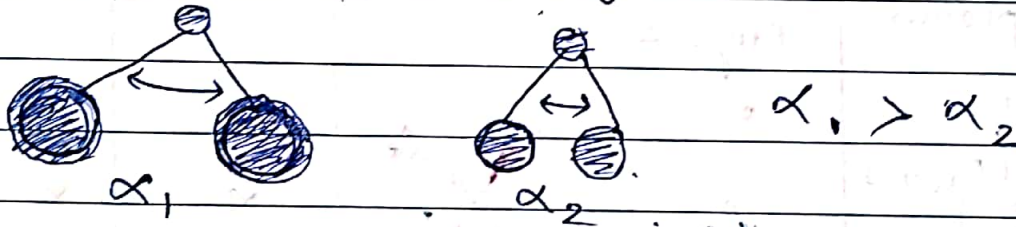
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role in deciding bond angles.

- Size of central atom $\propto$ bond angle



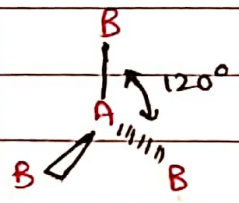
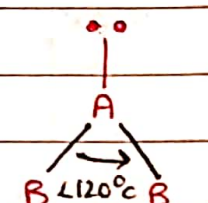
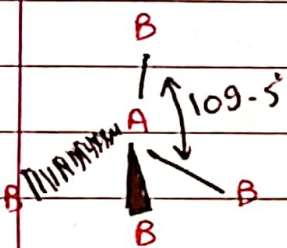
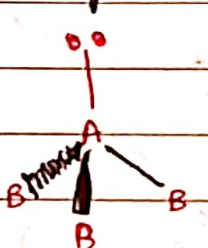
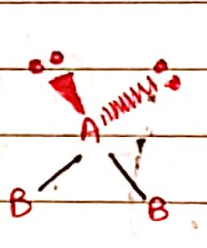
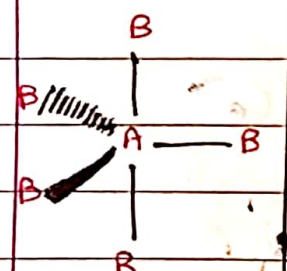
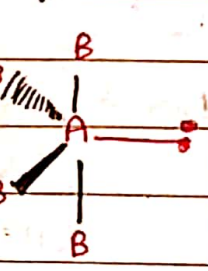
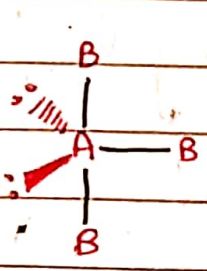
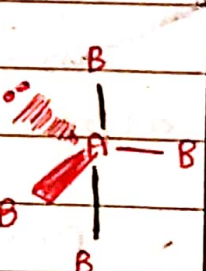
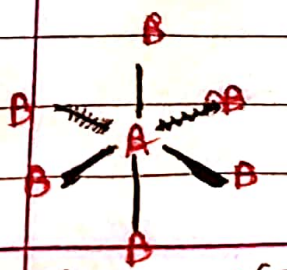
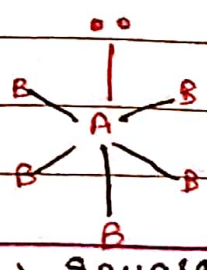
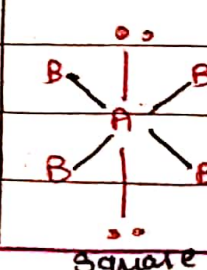
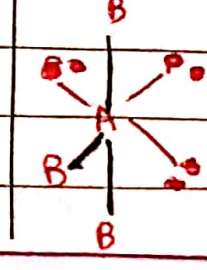
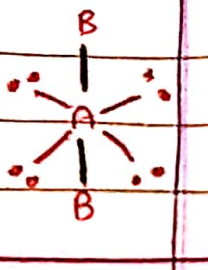
- Size of surrounding atom $\propto$ Bond angle



- Electronegativity of central atom $\propto$ Bond angle.

- Electronegativity of surrounding atom $\propto$ Bond angle

order of Bond angle $\Rightarrow \text{NH}_3 > \text{NF}_3 > \text{OH}_2 > \text{OF}_2$

steric Number	Basic geometry l.p = 0	L.P = 1	L.P = 2	L.P = 3	L.P = 4
2	 Linear	x	x	x	x
3	 trigonal planar	 Bent or Angular	x	x	x
4	 tetrahedral (Td)	 trigonal pyramidal	 angular or Bent	x	x
5	 Trigonal bipyramidal (TBP)	 See-Saw	 T-shaped	 Linear	x
6	 octahedral (Oh)	 square pyramidal	 square planar	 T-shaped	

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B.P

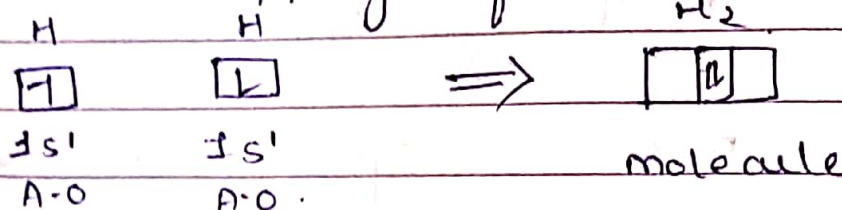
L.P

eg H_2O H_2O H_2O ; sp^3

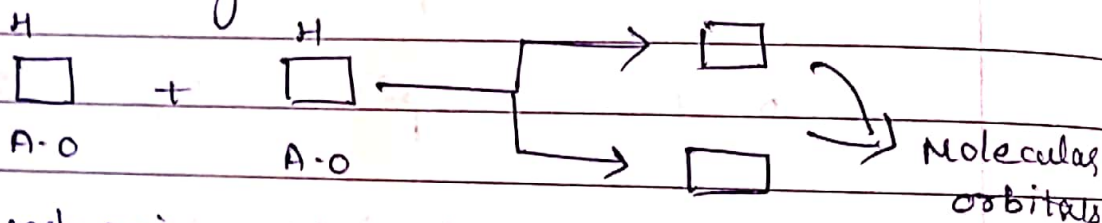
Molecule	Steric No.	Hyb ⁿ	B.P	L.P	geomet	Shape
H_2O	$n=4$	sp^3	2	2	Tetrahedral	Bent or Angular
SF_6	$n=6$	sp^3d^2	6	0	T.B.P	see saw
ClF_3	$n=5$	sp^3d	3	2	T.B.P	T Shape
XeO_3	$n=4$	sp^3	3	1	Td	Trigonal pyramidal
SnCl_2	$n=3$	sp^2	2	1	Trigonal planar	Bent
ICl_2^-	$n=5$	sp^3d	2	3	T.B.P	linear
XeF_4	$n=6$	sp^3d^2	4	2	Oh	Square planar
XeOF_4	$n=6$	sp^3d^2	5	1	Oh	Square pyramidal
XeF_6	$n=7$	sp^3d^3	6	1	P.B.P	distorted Octahedral
XeF_5^+	$n=6$	sp^3d^2	5	1	Oh	Square pyramidal
XeO_3F_2	$n=5$	sp^3d	5	0	T.B.P	T.B.P
I_3^-	$n=5$	sp^3d	2	3	T.B.P	linear

Molecular Orbital Theory (MOT)

According to VBT; a molecule is formed by overlapping of atomic orbitals.



But according to MOT:-



By overlapping of atomic orbitals new molecular orbitals are formed.

Important points of MOT.

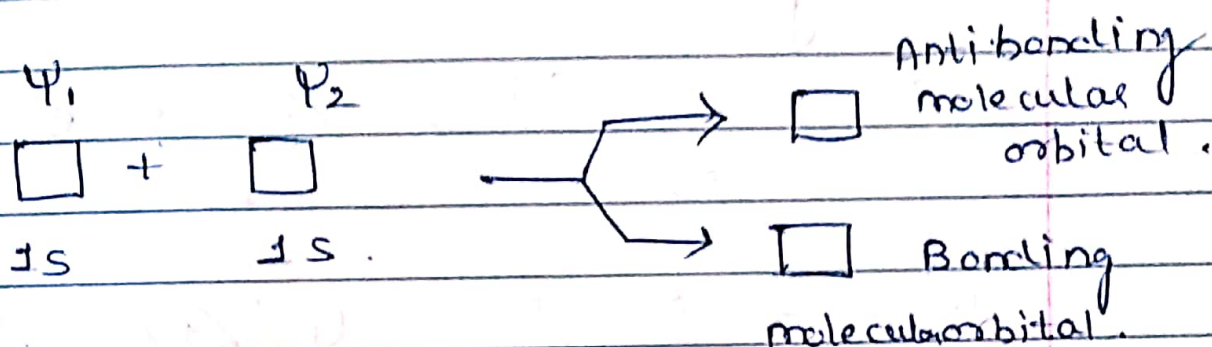
orbitals of atoms are called atomic orbitals when these orbitals combine to form molecule, the new orbitals of molecule are called molecular orbitals.

for writing electronic configuration of atom we use $1s, 2s, 2p, 3s, \dots$ notations

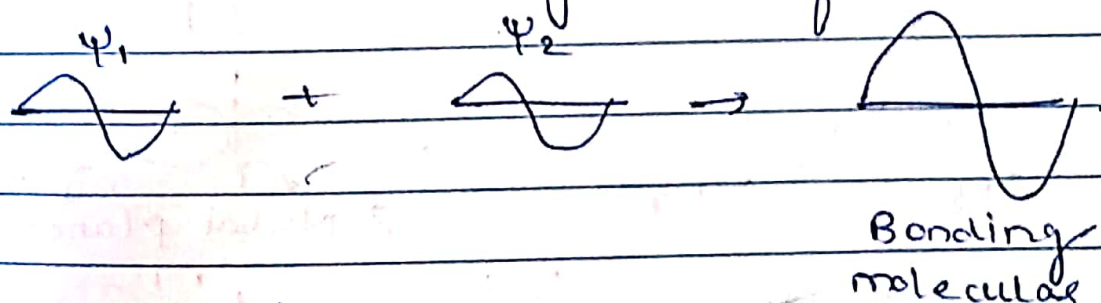
but for electronic configuration of molecules we use $\sigma/\pi/p/s...$

notations.

No. of molecular orbital formed are always equal to the no. of atomic orbitals combined.

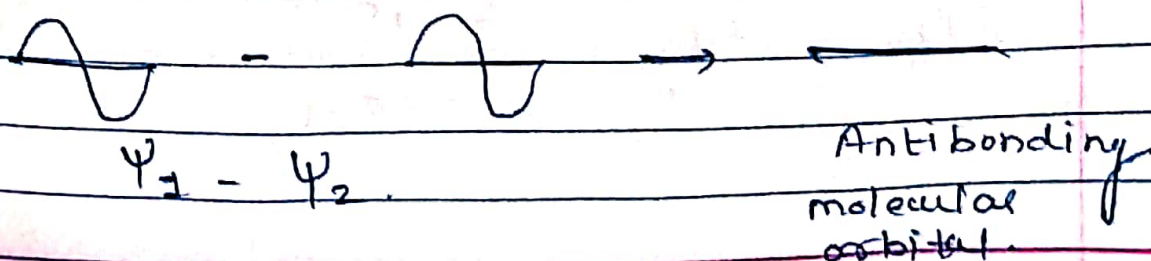


Bonding molecular orbital is formed by constructive interference of waves



Low in Energy. $\leftarrow$ orbital (stable)

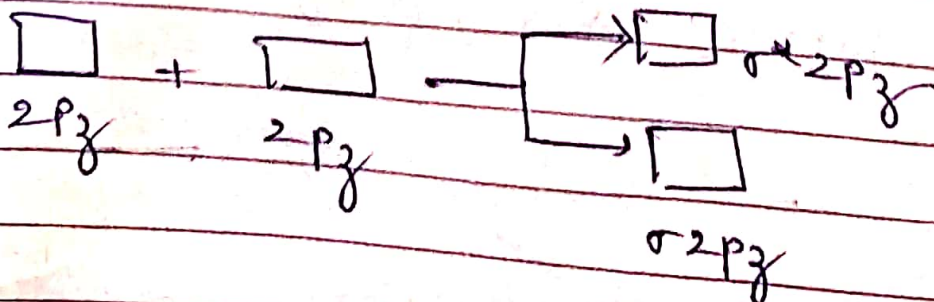
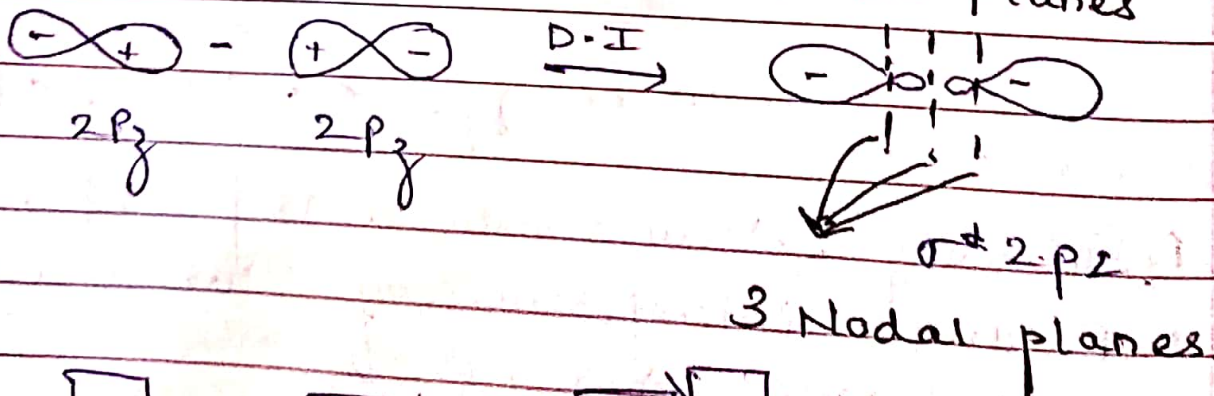
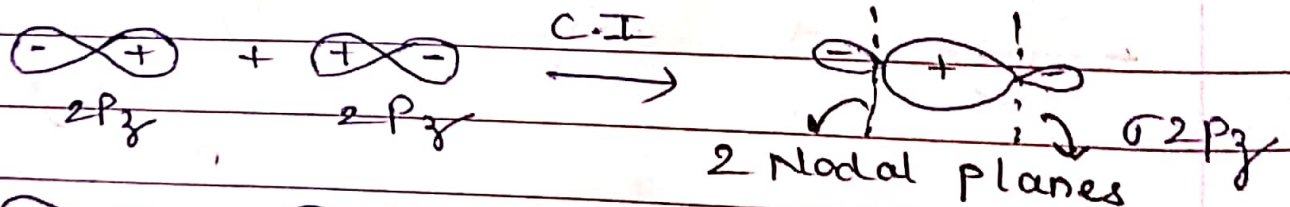
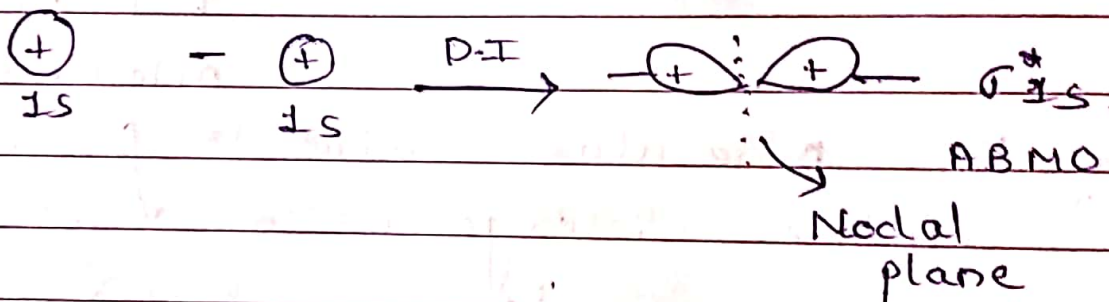
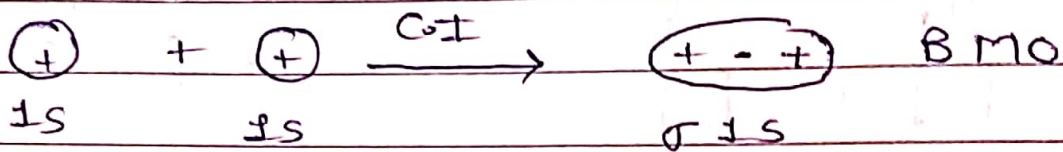
Antibonding molecular orbital is formed by destructive interference of atomic orbitals (waves)



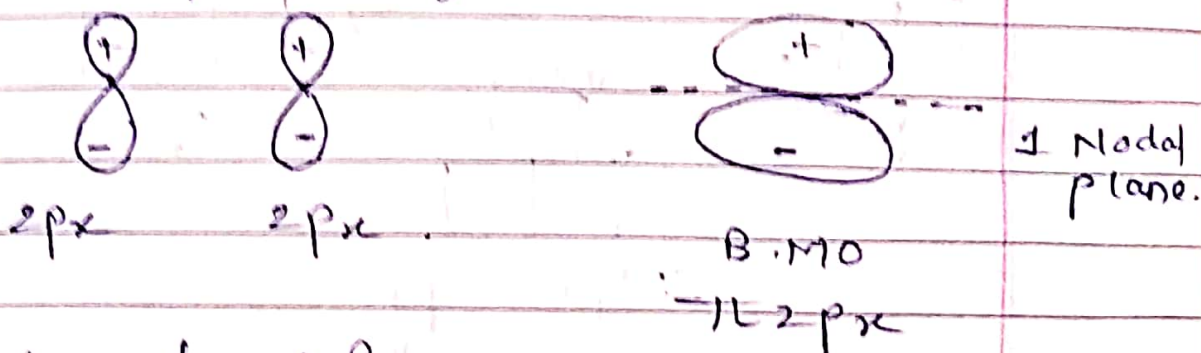
High in Energy. $\leftarrow$ unstable

Combining molecular orbitals have same or nearly same Energy and shape.

Molecular orbitals are formed by axial overlap or axial atomic orbital interference.



π molecular orbitals $\rightarrow$ Sidewise overlapping or interference of atomic orbitals.



Similarly for $2p_y$.

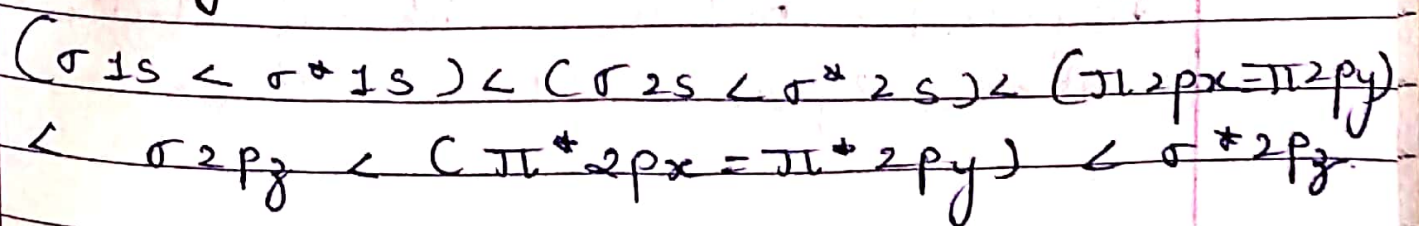
How to do electronic Configuration of molecule.

① Aufbau's principle $\rightarrow$ electron enters in increasing order of energy.

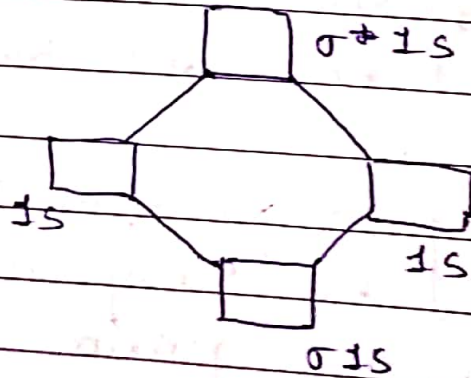
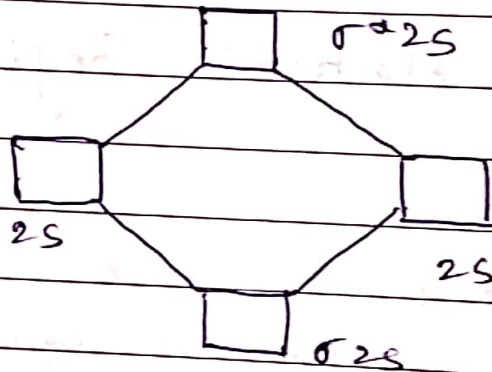
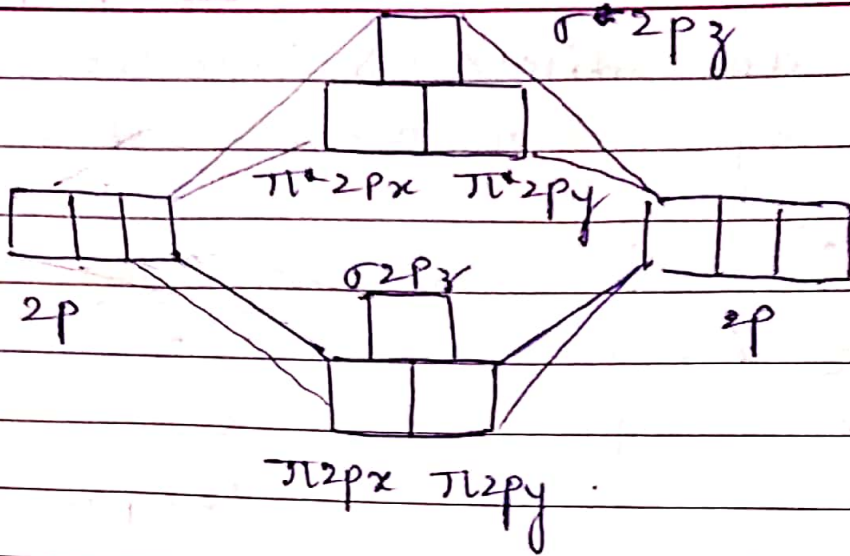
Pauli's principle $\rightarrow$ one orbital can have maximum two electrons and the spin must be opposite.

Hund's rule $\rightarrow$ if energy of orbitals are same then e^- will be unpaired and with same spin after that pairing starts.

Case I If we have less than or equal to $14e^-$.



ENERGY.



Molecular Orbital diagram/ Energy level diagram for molecules having electrons less than or equal to 14.

aq H_2 Configuration $\Rightarrow \sigma 1s^2$.

Bond order = $\frac{\text{no of } e^- \text{ in Bonding orbitals} - \text{no of } e^- \text{ in antibonding orbitals}}{2}$

$\therefore$ B.O of H_2 is $\frac{2-0}{2} = 1$.

He_2 $\sigma 1s^2 \sigma^* 1s^2$ $\rightarrow$

$$B.O = \frac{2-2}{2} = 0$$

$\therefore$ Bond order of He_2 is zero, it does not exist.

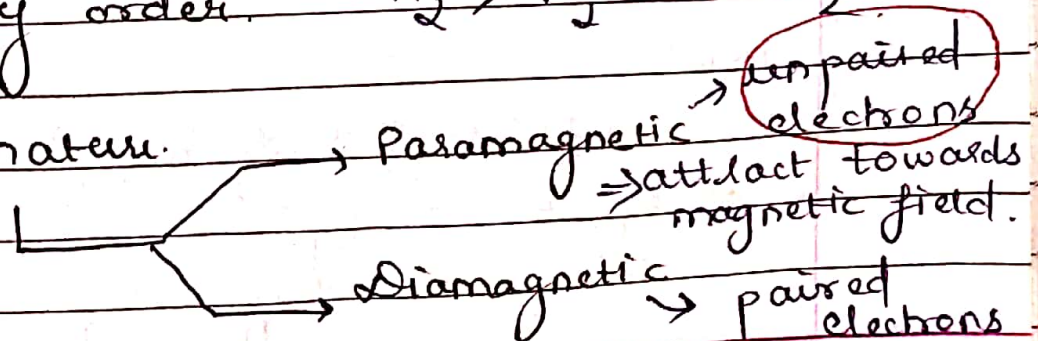
H_2^+ $\sigma 1s^1$

$$B.O = \frac{1-0}{2} = 0.5$$

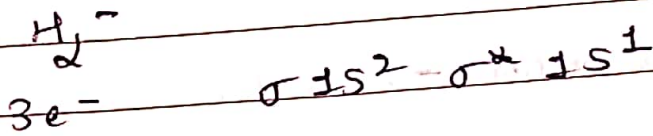
Stability of molecule $\propto$ Bond order

$\therefore$ Stability order: $H_2 > H_2^+ > He_2$

Magnetic nature.



• repelled by magnetic field.



$$B.O = \frac{2-1}{2} = \frac{1}{2}$$

stability order of H_2 , H_2^- , H_2^+ ?

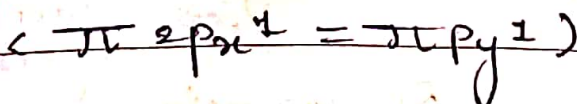
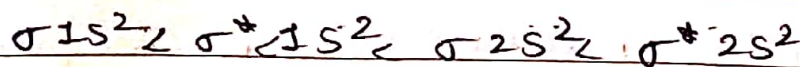
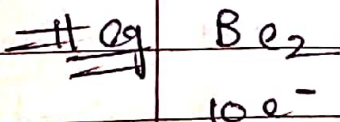
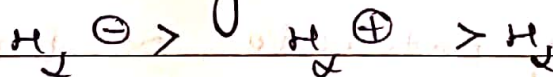
Since B.O. bond order of H_2^+ and H_2^- are equal.

In such cases $\Rightarrow$ where B.O. of two molecules are equal, then molecules having lesser number of electrons in antibonding molecular orbital is more stable

$\therefore$ stability order $\text{H}_2 > \text{H}_2^+ > \text{H}_2^-$

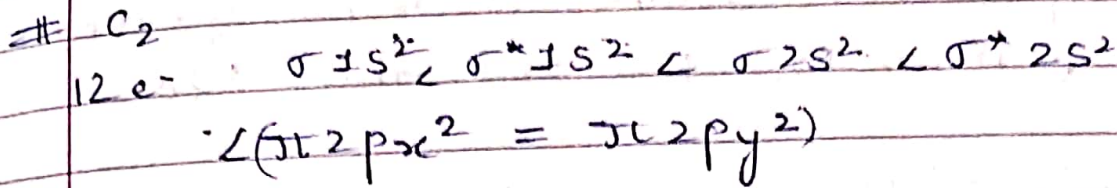
$\#$ Bond length $\propto \frac{1}{\text{Bond order}}$

order of Bond length



$$B.O = \frac{6-4}{2} = \frac{2}{2} = 1$$

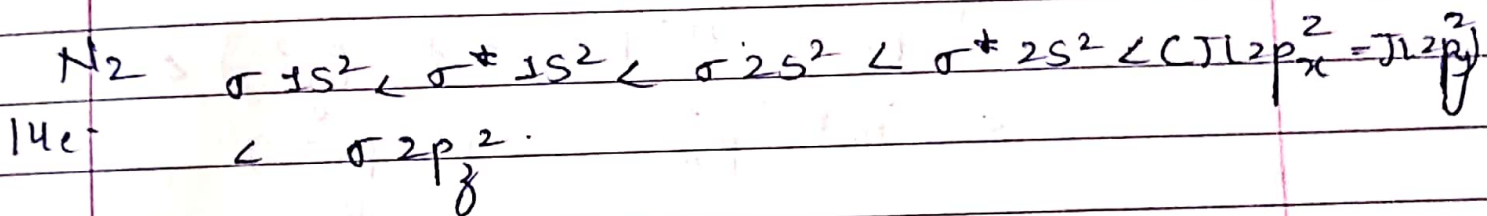
Magnetic nature $\Rightarrow$ paramagnetic



$$\Rightarrow B.O = \frac{8-4}{2} = \frac{4}{2} = 2$$

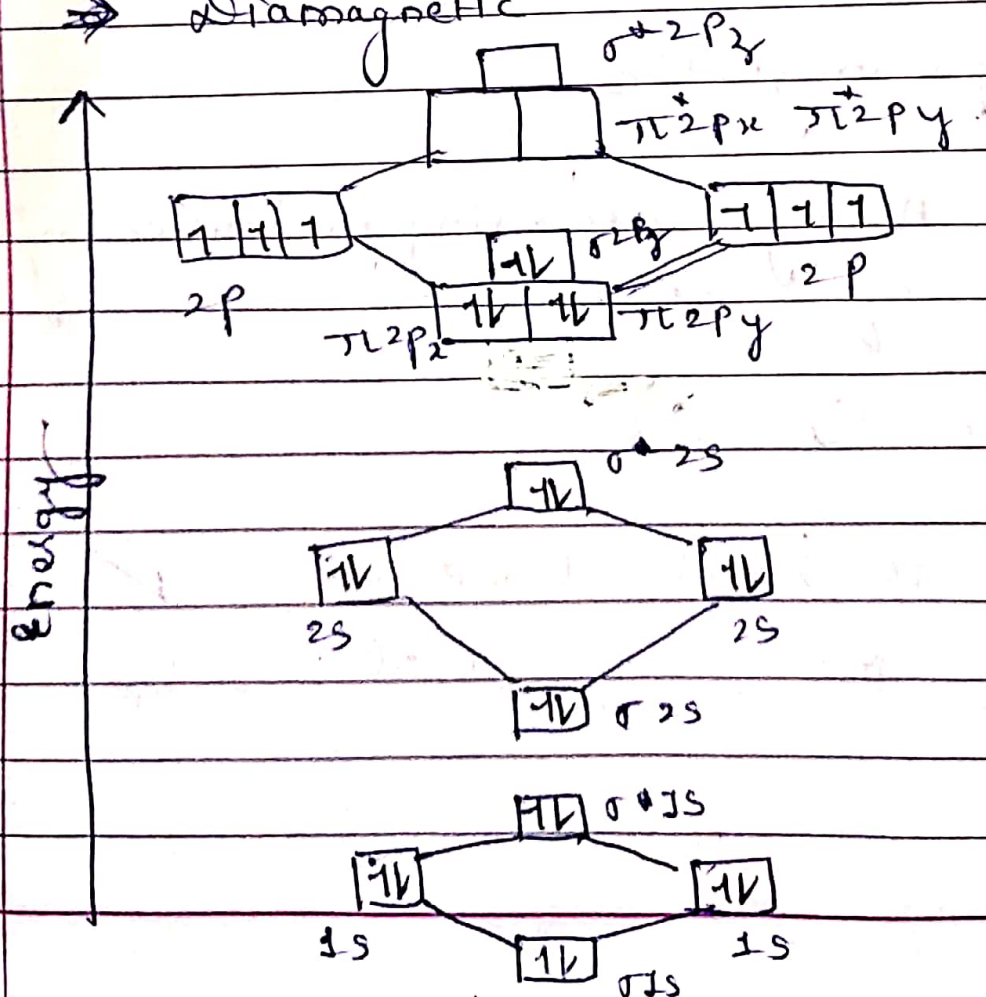
$\Rightarrow$ Diamagnetic

write down electronic configuration for N_2 and draw energy level / M.O diagram.



$$\Rightarrow B.O = \frac{10-4}{2} = \frac{6}{2} = 3$$

$\Rightarrow$ Diamagnetic



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Case II if molecule have electrons greater than 15 but less than 20.

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$$

Eg

O_2
16e-

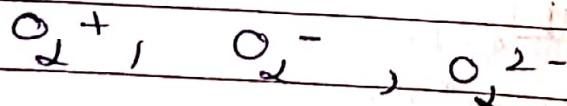
$$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^2 < (\pi 2p_x^2 = \pi 2p_y^2) < (\pi^* 2p_x^1 = \pi^* 2p_y^1)$$

$$B.O = \frac{10 - 6}{2} = 2$$

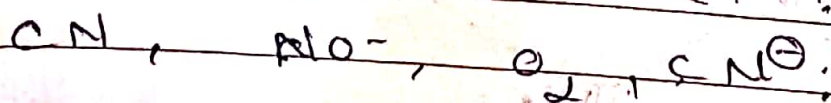
$\Rightarrow$ paramagnetic

Ans

which Arrange according to increasing order of stability

Ans

which is least stable?

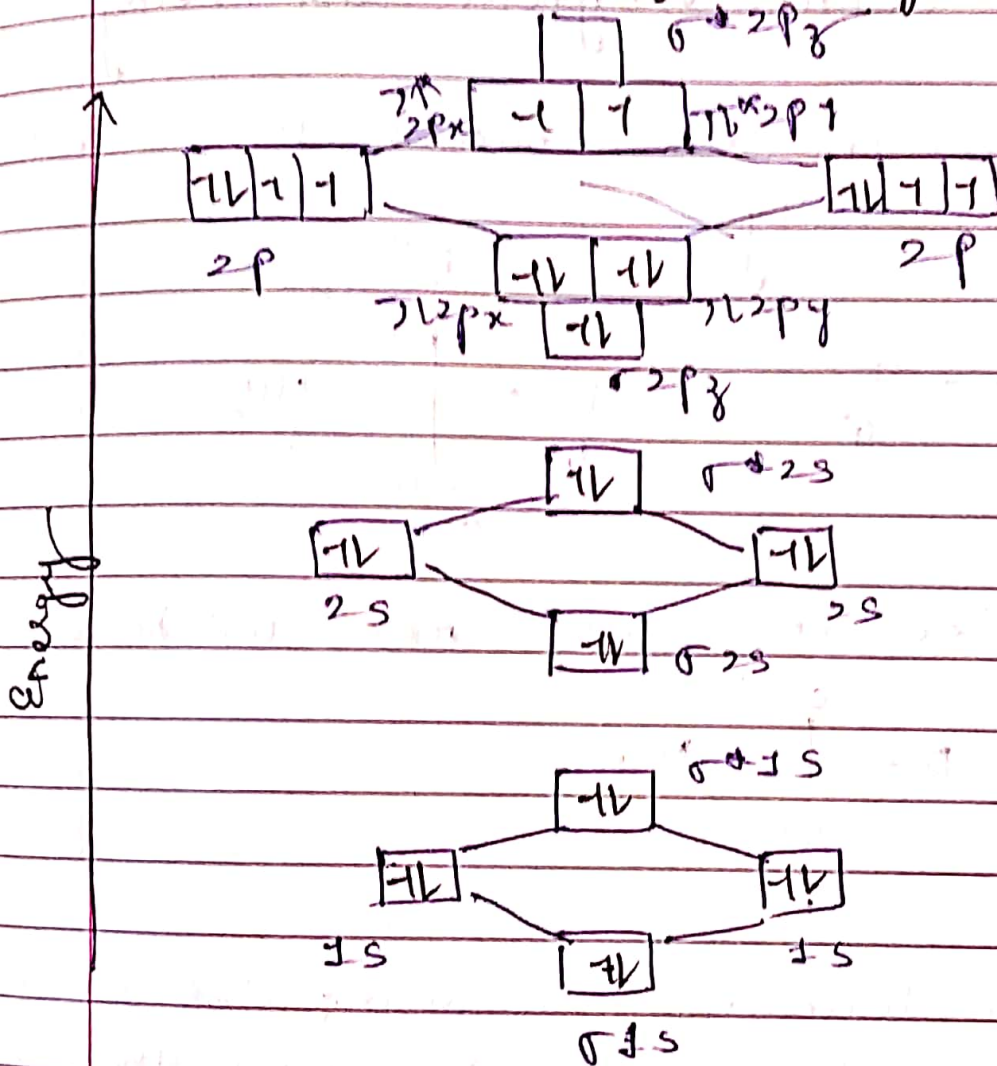


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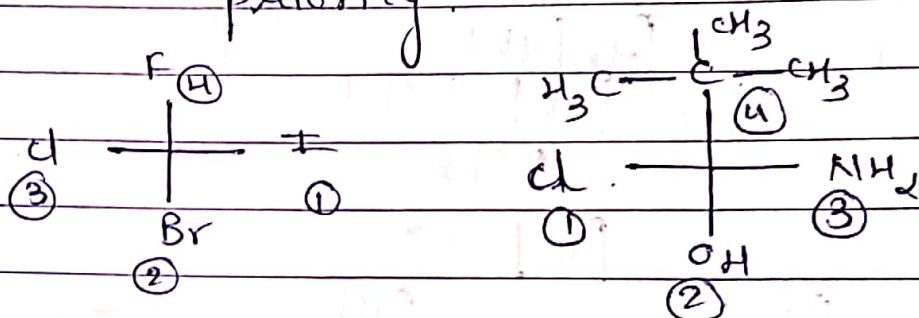
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Energy level diagram/M.O diagram for O_2

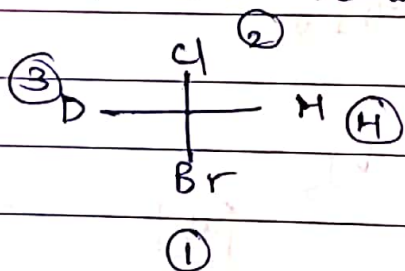


CIP Rule / Priority Sequence rule:-
given by Cahn-Ingold and Prelog.

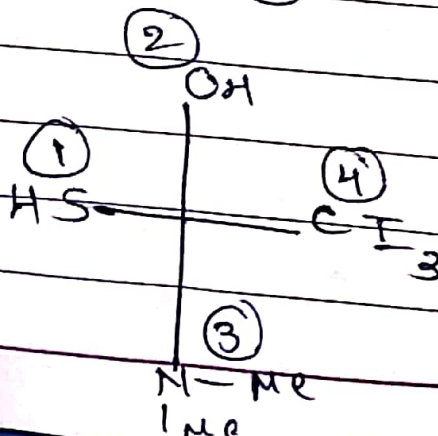
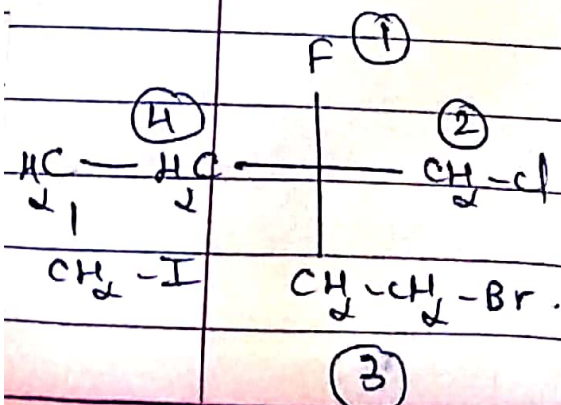
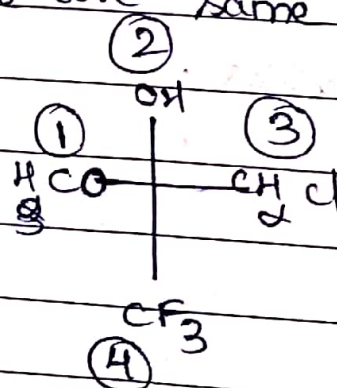
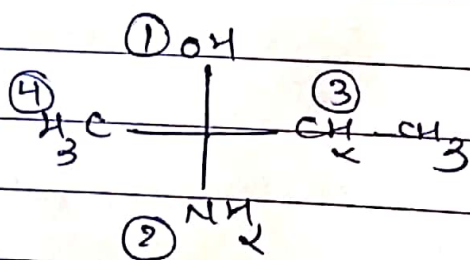
Rule 1) Higher the atomic number higher will be priority.



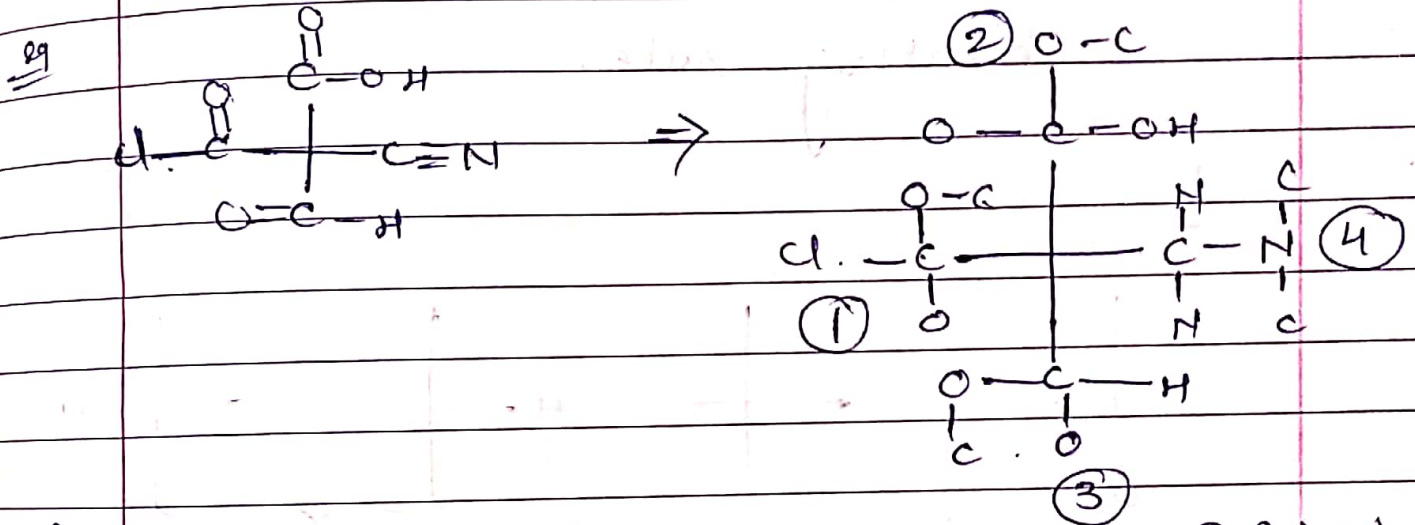
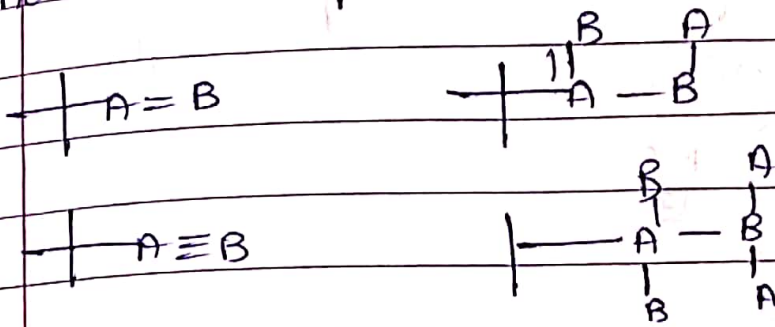
Rule 2) Heavier the isotope more will be priority.



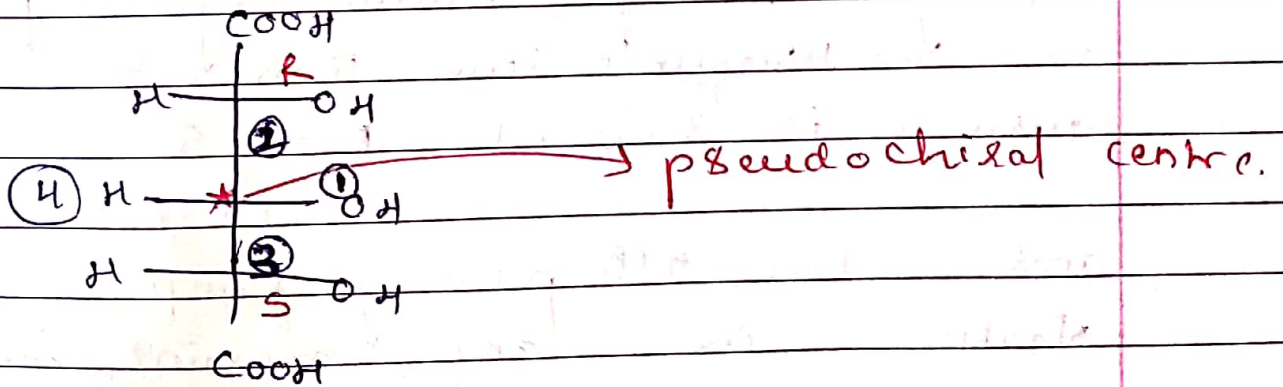
Rule 3) Move towards next atom; when atoms directly attached to chiral centre are same then see next atom attached to these atoms.



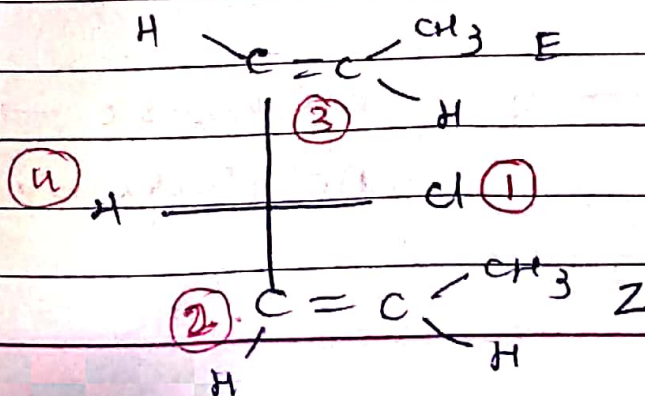
Rule no. 4:- Multiple bond are treated as follows:-



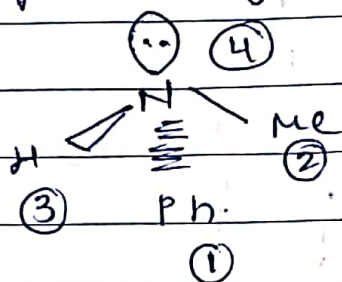
Rule no. 5:- R is dominant over S (R > S)



Rule no. 6:- Z is dominant over E (Z > E)



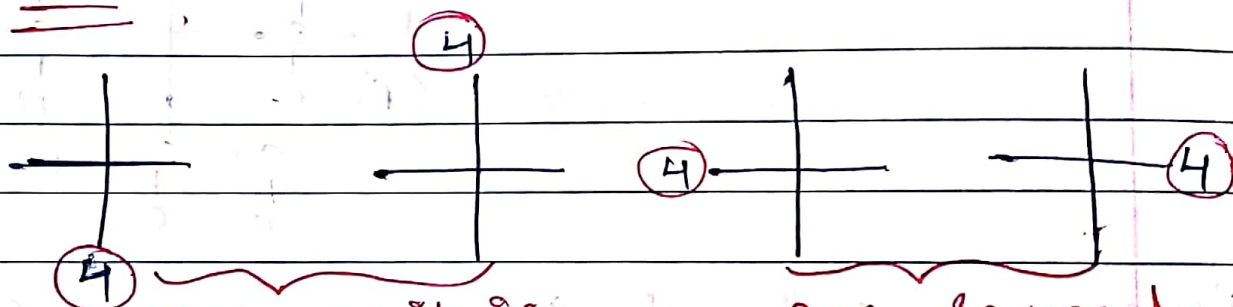
Rule 7 lone pair ^{cross} gets lowest priority.



Chirality rule.

①

Fischer



Ans. as it is

Ans. reversed

In fischer representation if 4th priority is on horizontal line then answer is reversed $R \rightarrow S$

$S \rightarrow R$

and when 4th priority group is on vertical line ans remains same

$R \rightarrow R$

$S \rightarrow S$

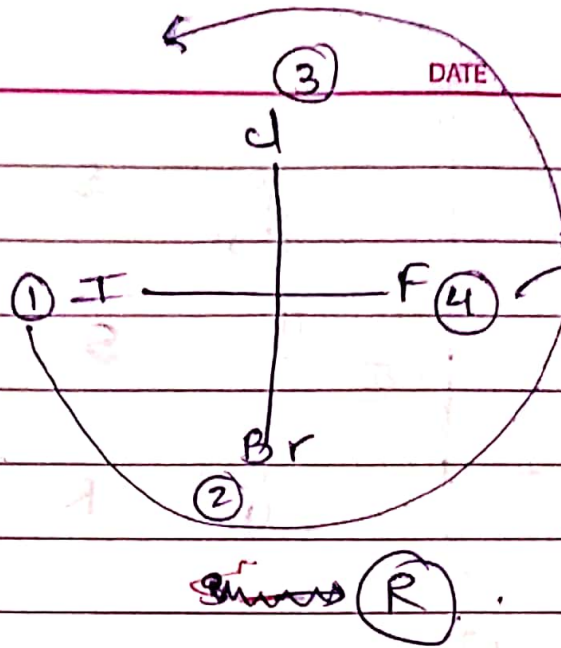
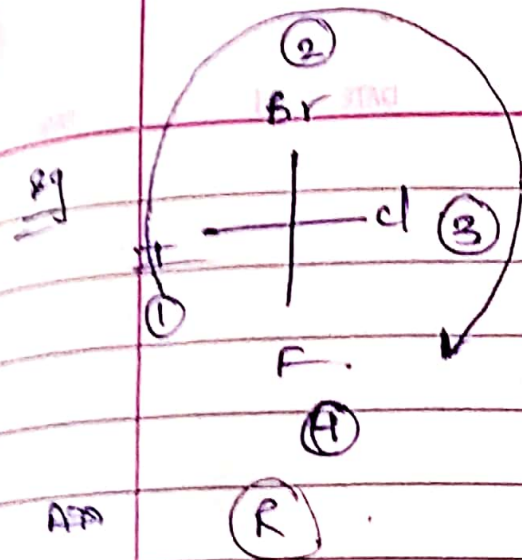
$1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ clockwise $\rightarrow R$

Anticlockwise $\rightarrow S$

25

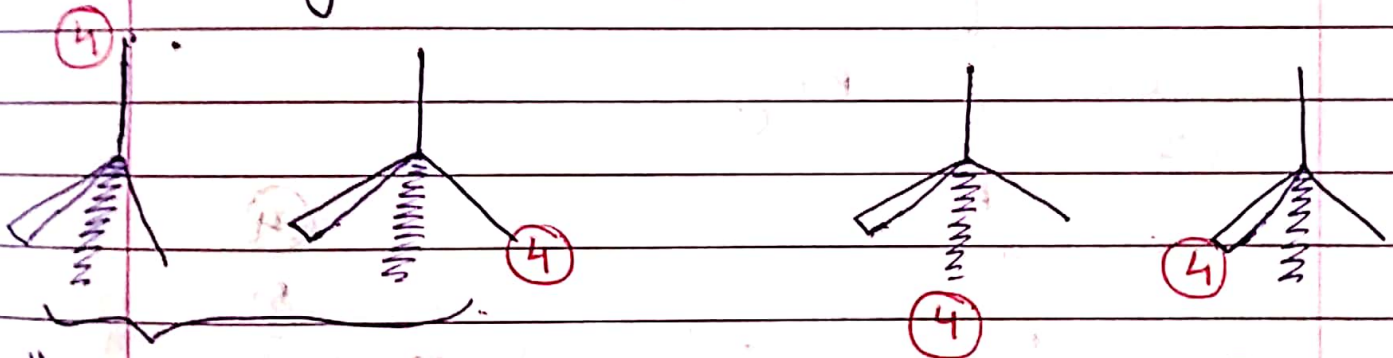
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since 4th priority on horizontal line, ans is R but it will be reversed i.e R.

2) Wedge dash representation.

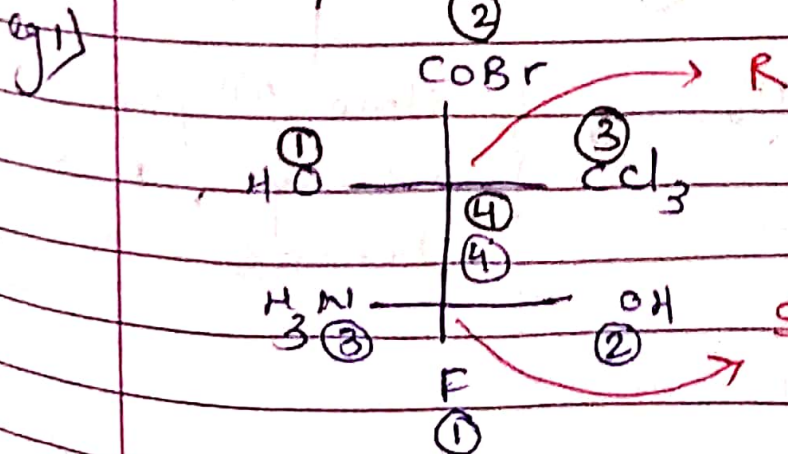


4th priority on this lines have to do even interchange.

Ans, will be as it is

Reverse your Answer

Examples:-



26

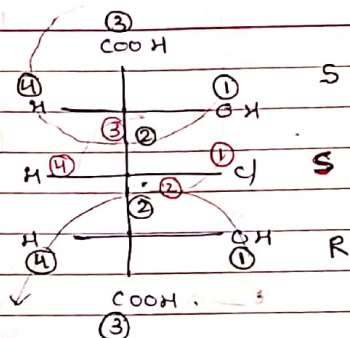
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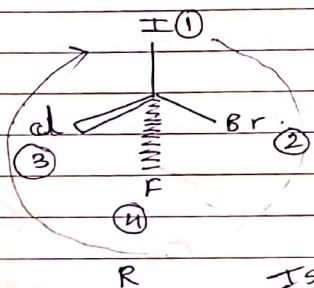
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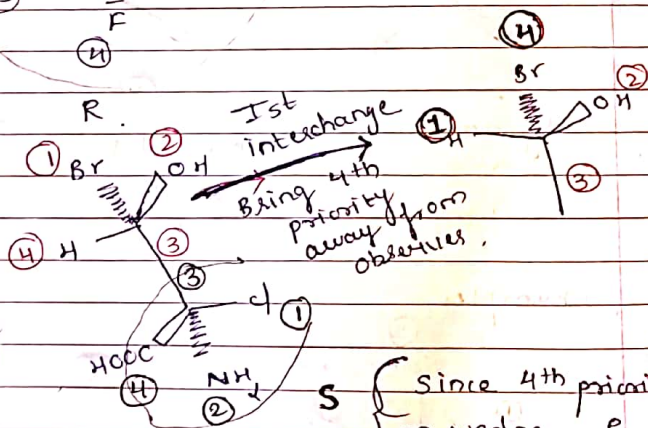
eg 2



eg 3



eg 4



4th priority
on plane of
paper → we
will have to
do even time

Interchange; so we will
interchange two times

Since 4th priority
on wedge, R
becomes S

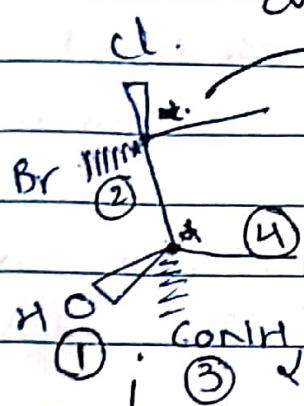
eg 5

Redrawing this
chiral center

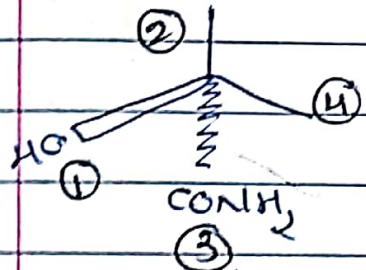
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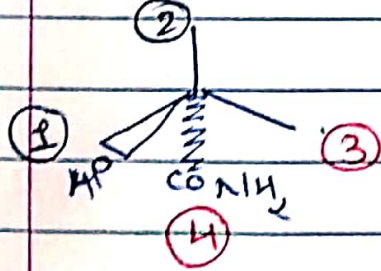
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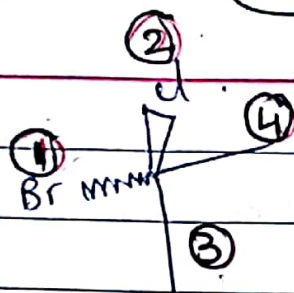
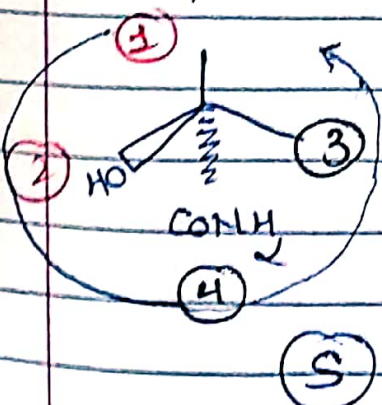
Redrawing
this chiral
center..



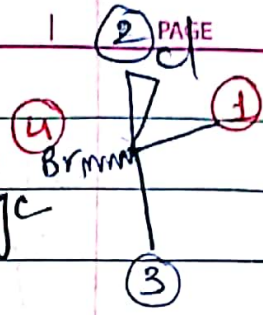
1st
Interchange



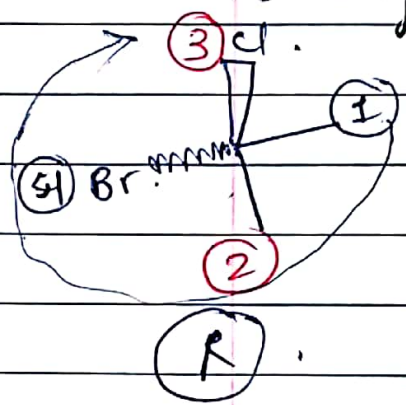
2nd Interchange.



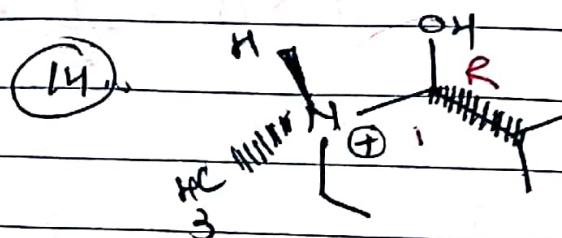
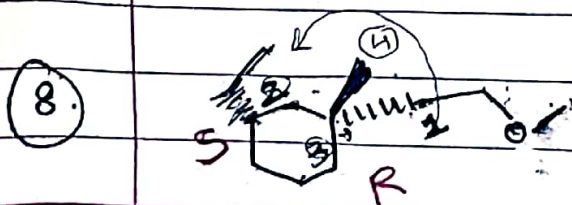
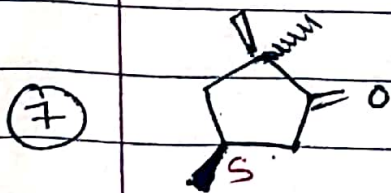
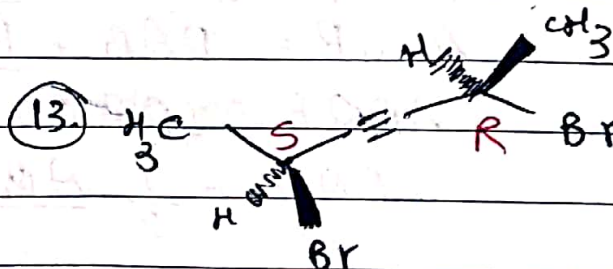
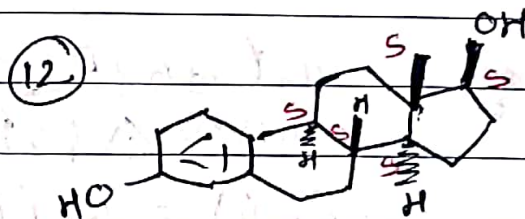
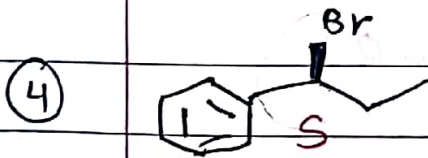
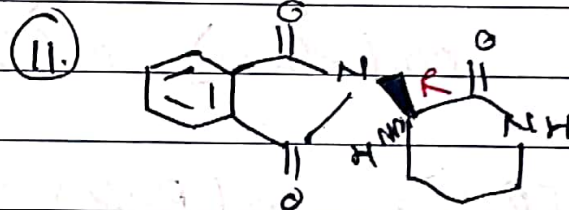
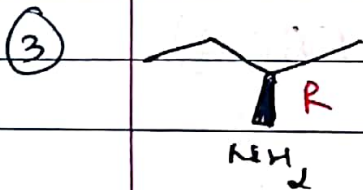
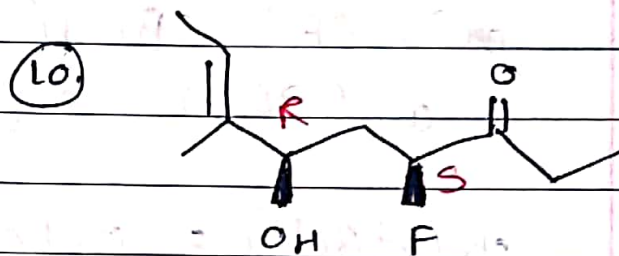
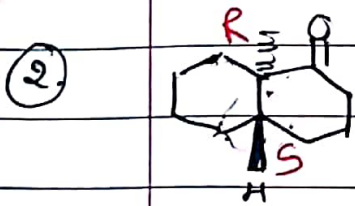
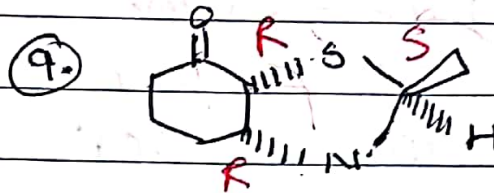
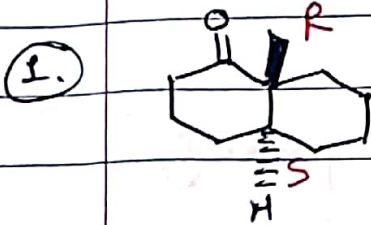
Interchange



II
Inter
change.



Que Find out the chiral Centres and assign absolute Configuration.

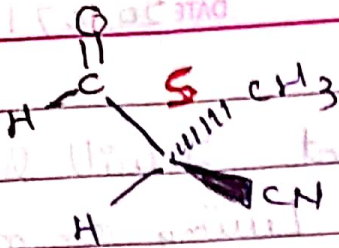


(29)

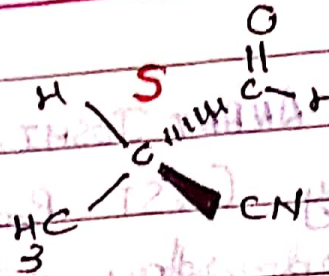
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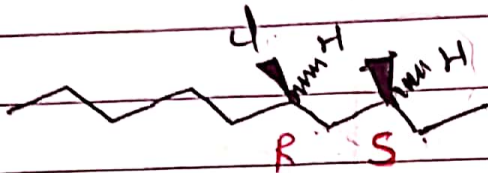
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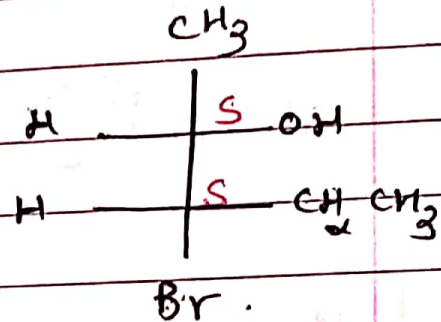
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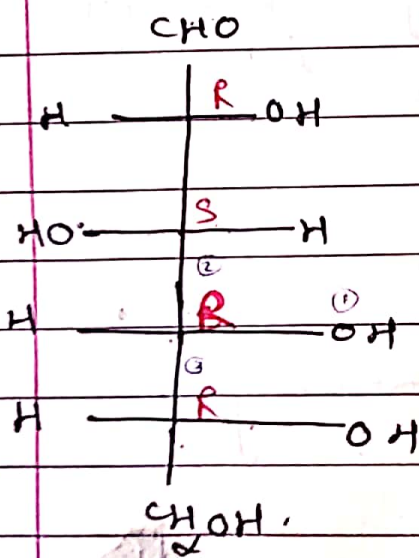
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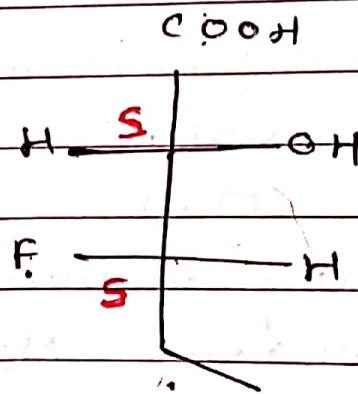
18



19



20



Requirement

1) Rotation about the bond is restricted.

→ π bond (Alkenes)

→ by incorporation of bond in a ring system.

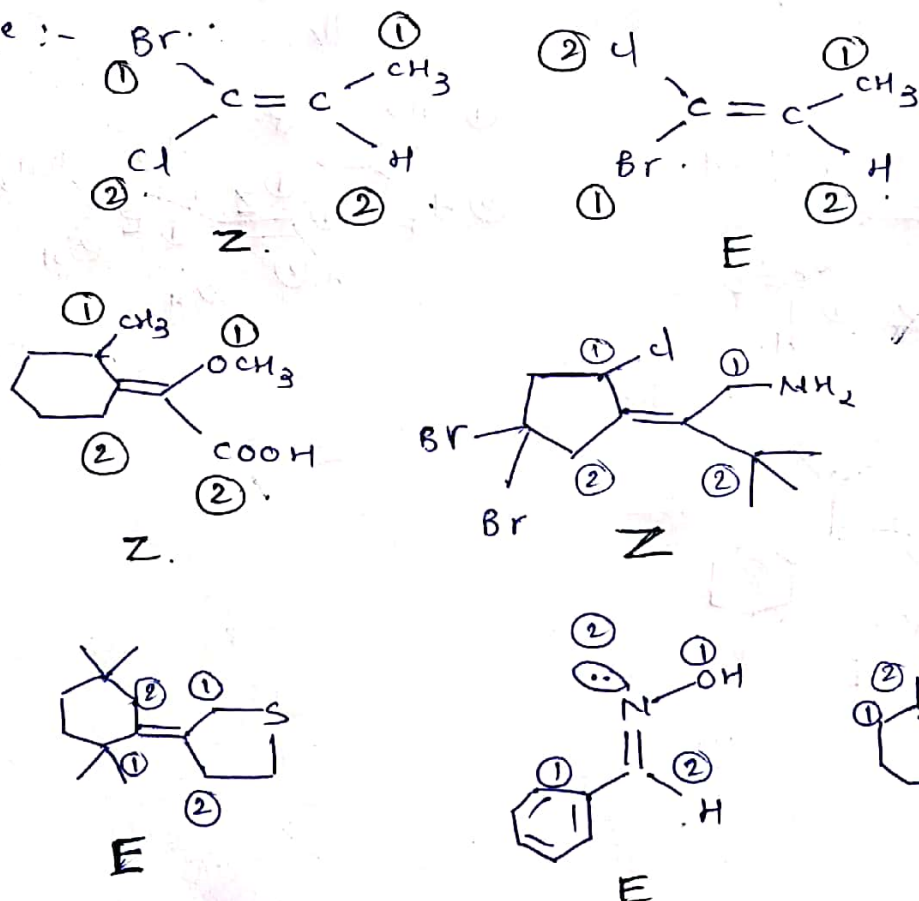
E-Z System of Nomenclature of alkenes

* Based on the priorities of groups; on the CIP rule. It is used to designate diastereomeric alkenes.

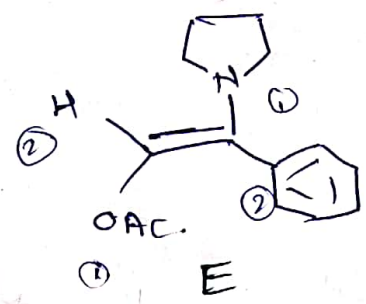
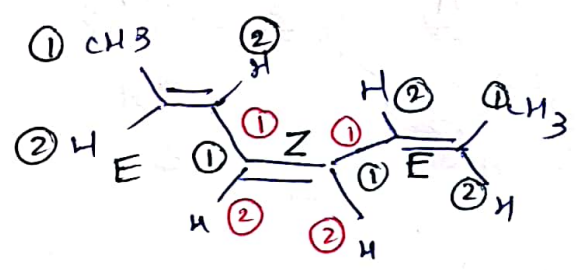
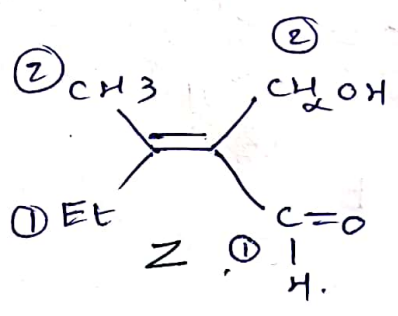
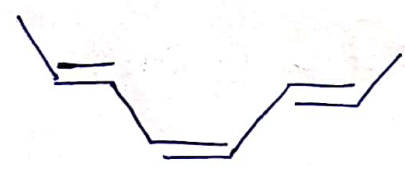
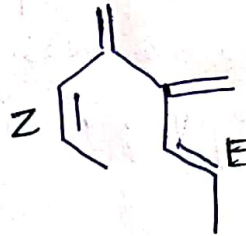
* In E-Z system 2 groups attached to each carbon of the double bond are arranged in order of priority.
eg $\text{Br} > \text{Cl} > \text{CH}_3 > \text{H}$.

* If the 2 groups of higher priority are located on the same side of double bond the alkene is designated **Z** isomer (from the German word *zusammen* meaning 'together') if on the other hand, 2 groups of higher priority are on opposite side of the double bond, the alkene is designated **E** isomer (German word *Entgegen* meaning 'opposite')

Example :-



3



21 32
 ≠ Molarity; The molarity, M , of a solution is the number of moles of solute in a litre of solution.

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{Volume of solution in litre}}$$

its unit is mol/litre or M .

Molality: no. of moles of solute in 1000 gm of solvent

$$\text{Molality} = \frac{\text{moles of solute}}{\text{wt. of solvent in kg}}; \text{ its unit is mol/kg or } m.$$

Normality: The Normality N , of a solution is the number of g equivalents i.e. equivalent weight expressed in grams of solute in a litre of solution.

$$N = \frac{\text{g equivalent of solute}}{\text{Volume of soln in litre}}$$

$$\text{No. of gm Equivalent} = \frac{W}{M/V} = \frac{\text{Mass}}{\text{Eq. Mass}} \left\{ \begin{array}{l} \text{Eq. Mass} = \frac{\text{Molar mass}}{\text{Valency factor}} \end{array} \right.$$

unit: gm eq/L, N .

Questions:-

1) 49 grams of H_2SO_4 are present in 100ml aqueous soln. what is the molarity of H_2SO_4 .

Soln $\rightarrow$ $W = 49 \text{ gm}$, $V = 100 \text{ ml} = 0.1 \text{ litre}$, $m \text{ of } H_2SO_4 = 2 + 32 + 16 \times 4 = 98$

$$M = W / M V (L)$$

$$= 49 / (98 \times 0.1)$$

$$M = 5 \text{ mole/litre}$$

2) An aq. solution of HCl is 38% by mass and its density is 1.19 gm/ml . Calculate the molality and molarity of solution.

(33)

Soln → Mass of solute 'w' = 38 gm

Mass of solution = 100 gm

Mass of solvent 'W' = $100 - 38$

= 62 gm = 0.062 kg ; m of HCl = $1 + 35.5 = 36.5$

molality = $w / m \times W(\text{kg})$

$$= 38 / (36.5 \times 0.062)$$

molality = 16.79 mole/kg

$M = w / m \times V(\text{L})$

$$38 / (36.5 \times 0.084)$$

Molarity = 12.39 mole/liter

Ques The density of a 2.03 M solution of acetic acid (molecular weight = 60) in water is 1.027 gm/ml . calculate Molality of solution.

Soln → $S =$

Ans A solution contains 410.3 grams of H_2SO_4 per liter of solution at 20°C . If the density is 1.243 gm/ml , what will be its Molarity and Molality.

$w = 410.3 \text{ gm}$

$V = 1 \text{ L}$

$m = 2 \times 1 + 32 + 16 \times 4 = 98$

Molarity = $w / m \times V(\text{L})$

$$= 410.3 / 98 \times 1$$

Molarity = 4.187 mole/liter

Mass of solution = Volume of Soln $\times$ density

$$= 1000 \times 1.243$$

$$= 1243 \text{ gm}$$

weight of solvent = $1243 - 410.3 = 832.7 \text{ gm}$
= 0.8327 kg.

Molality = $w / m \times W(\text{kg})$

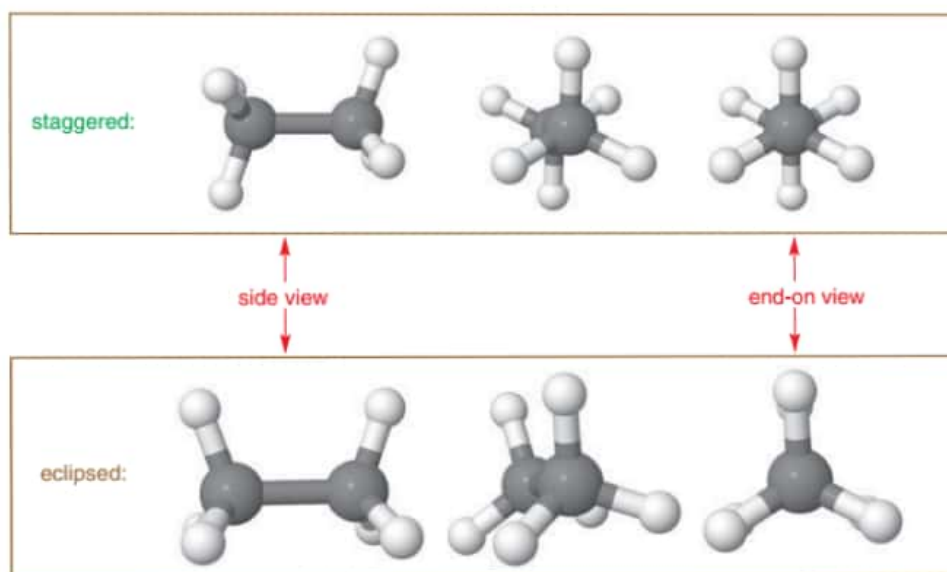
$$= 410.3 / 98 \times 0.8327$$

Molality = 5.028 mole/kg

Conformations of ethane

Why should there be an energy barrier in the rotation about a single bond? In order to answer this question, we should start with the simplest C—C bond possible—the one in ethane. Ethane has two extreme conformations called the staggered and eclipsed conformations. Three different views of these are shown below.

the two extreme conformations of ethane, staggered and eclipsed, each shown from three different viewpoints

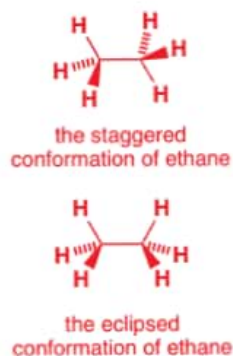


You can see why the conformations have these names by looking at the end-on views in the diagram. In the eclipsed case the near C—H bonds completely block the view of the far bonds, just as in a solar eclipse the Moon blocks the Sun as seen from the Earth. In the staggered conformation, the far C—H bonds appear in the gaps between the near C—H bonds—the bonds are staggered.

Chemists often want to draw these two conformations quickly and two different methods are commonly used, each with its own merits. In the first method, shown on the right, we simply draw the side view of the molecule and use wedged and hashed lines to show bonds not in the plane of the paper (as you saw in Chapter 14). Particular attention must be paid to which of the bonds are in the plane and which go into and out of the plane.

In the second method we draw the end-on view, looking along the C—C bond. This view is known as a Newman projection, and Newman projections are subject to a few conventions:

- The carbon atom nearer the viewer is at the junction of the front three bonds.
- The carbon further away (which can't in fact be seen in the end-on view) is represented by a large circle. This makes the perspective inaccurate, but this doesn't matter.



- Bonds attached to this further carbon join the *edge* of the circle and do not meet in the centre.
- Eclipsed bonds are drawn slightly displaced for clarity—as though the bond were rotated by a tiny fraction.

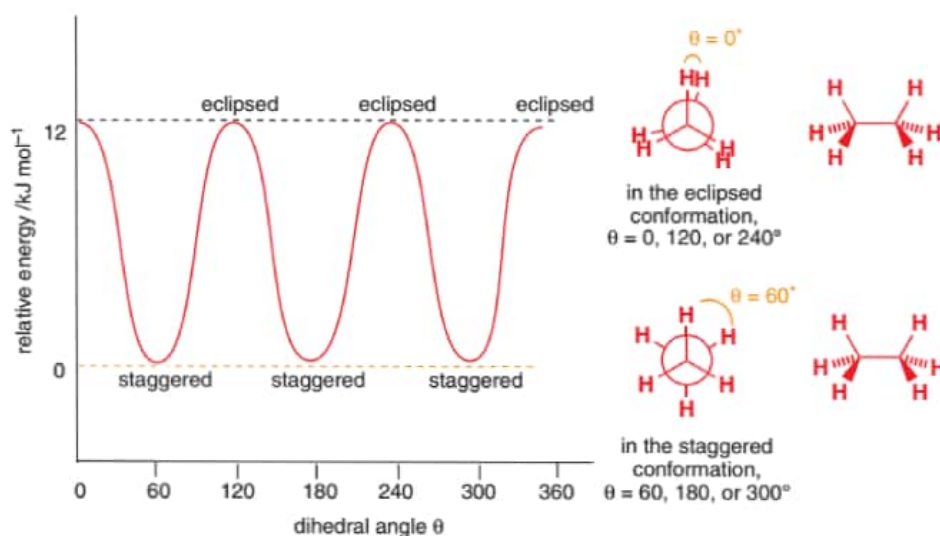
Newman projections of the staggered and eclipsed conformations of ethane



■ Picturing dihedral angles is sometimes hard—one way to do it is to imagine the two C–H bonds drawn on two facing pages of a book. The dihedral angle is then the angle between the pages, measured perpendicular to the spine. See Chapter 31.

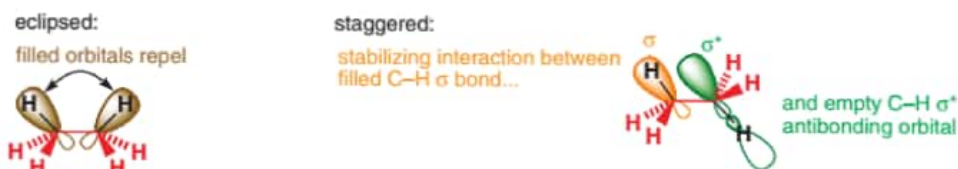
The staggered and eclipsed conformations of ethane are not identical in energy: the staggered conformation is lower in energy than the eclipsed by 12 kJ mol^{-1} , the value of the rotational barrier. Of course, there are other possible conformations too with energies in between these extremes, and we can plot a graph to show the change in energy of the system as the C–C bond rotates. We define the dihedral angle, θ (sometimes called the torsion angle), to be the angle between a C–H bond at the nearer carbon and a C–H bond at the far carbon. In the staggered conformation, $\theta = 60^\circ$ whilst in the eclipsed conformation, $\theta = 0^\circ$.

The energy level diagram shows the staggered conformation as a potential energy minimum whilst the eclipsed conformation represents an energy maximum. This means that the eclipsed conformation is not a stable conformation since any slight rotation will lead to a conformation lower in energy. The molecule will actually spend the vast majority of its time in a staggered or nearly staggered conformation and only briefly pass through the eclipsed conformation *en route* to another staggered conformation.



Interactive conformations of ethane

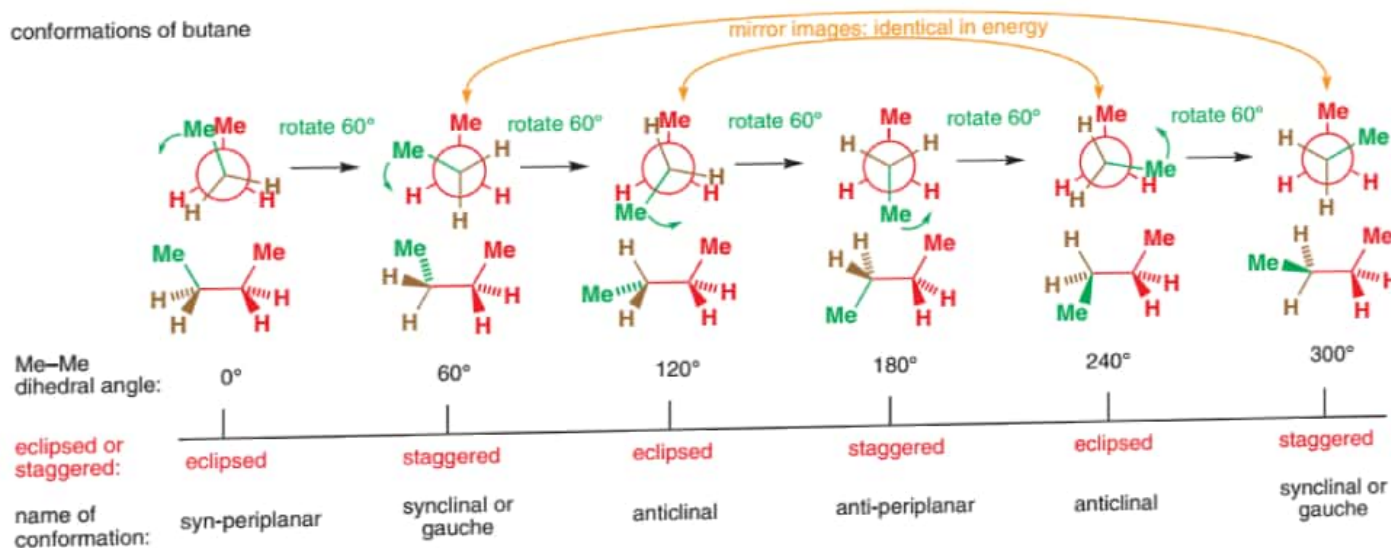
But *why* is the eclipsed conformation higher in energy than the staggered conformation? There are two reasons. The first is that the electrons in the bonds repel each other and this repulsion is at a maximum when the bonds are aligned in the eclipsed conformation. The second is that there may be some stabilizing interaction between the C–H σ bonding orbital on one carbon and the C–H σ^* antibonding orbital on the other carbon, which is greatest when the two orbitals are exactly parallel: this only happens in the staggered conformation. The same effects—repulsion between filled orbitals (a form of steric effect, see p. 129) and stabilization by donation into antibonding orbitals—govern the favoured conformations about all rotating bonds.



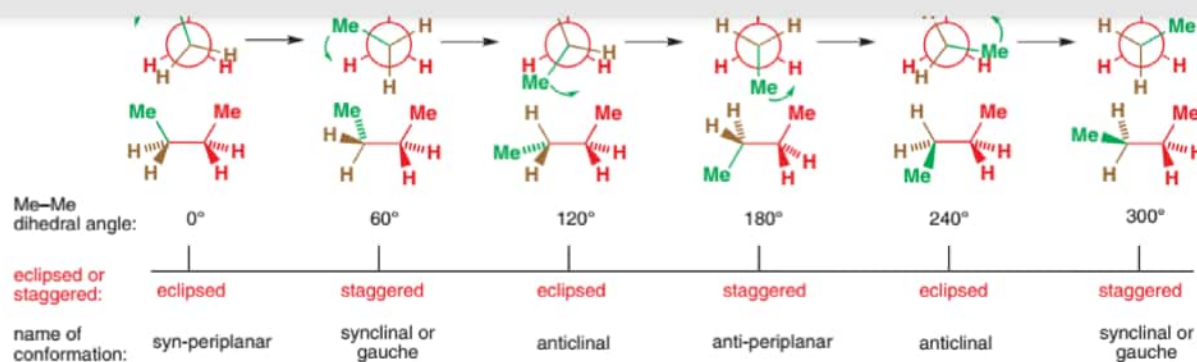
Conformations of butane

With butane things start to get slightly more complicated. Now we have effectively replaced two hydrogen atoms in ethane by larger methyl groups. These are large enough to get in the way of each other, and steric repulsion becomes a significant contribution to the rotational energy barriers. However, the main complication is that, as we rotate about the central C–C bond, not all the staggered conformations are the same, and neither are all the eclipsed conformations. The six conformations that butane can adopt as the central C–C bond is rotated in 60° intervals are shown below. The green Me group and the brown hydrogens are rotating while the substituents on the other carbon atom remain still.

conformations of butane

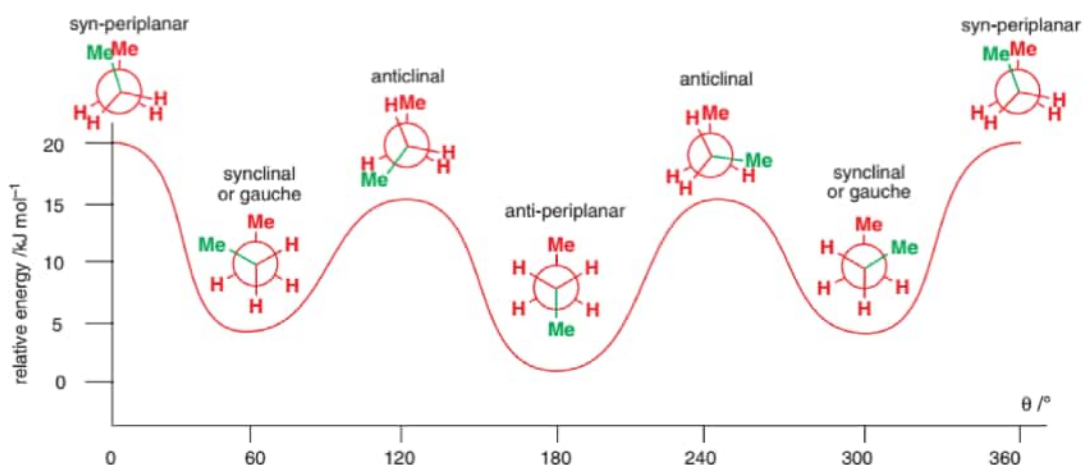


Look closely at these different conformations. The conformations with dihedral angles 60° and 300° are actually mirror images of each other, as are the conformations with angles 120° and 240° . This means that we really only have four different maxima or minima in energy as we rotate about the central C–C bond: two types of eclipsed conformation, which will represent maxima in the energy-rotation graph, and two types of staggered conformation, which will represent minima. These four different conformations have names, shown in the bottom row of the diagram. In the syn-periplanar and anti-periplanar conformations the two C–Me bonds lie in the same plane; in the synclinal (or gauche) and anticlinal conformations they slope towards (*syn*) or away from (*anti*) one another.



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Before we draw the energy–rotation graph, let's just stop and think what it might look like. Each of the eclipsed conformations will be energy maxima but the syn-periplanar conformation ($\theta = 0^\circ$) will be higher in energy than the two anticlinal conformations ($\theta = 120^\circ$ and 240°): in the syn-periplanar conformation two methyl groups are eclipsing each other whereas in the anticlinal conformations each methyl group is eclipsing only a hydrogen atom. The staggered conformations will be energy minima but the two methyl groups are furthest from each other in the anti-periplanar conformation so this will be a slightly lower minimum than the two synclinal (gauche) conformations.



Interactive conformations of butane

■ The rotation is very rapid indeed: the barrier of 20 kJ mol^{−1} corresponds to a rate at room temperature of 2×10^9 s^{−1}. This is far too fast for the different conformers to be detected by NMR (see p. 363);

As in ethane, the eclipsed conformations are not stable since any rotation leads to a more stable conformation. The staggered conformations are stable since they each lie in a potential energy well. The anti-periplanar conformation, with the two methyl groups opposite each other, is the most stable of all. We can therefore think of a butane molecule as rapidly interconverting between synclinal and anti-periplanar conformations, passing quickly through the eclipsed conformations on the way. The eclipsed conformations are energy maxima, and therefore represent the transition states for interconversion between conformations.

If we managed to slow down the rapid interconversions in butane (by cooling to very low temperature, for example), we would be able to isolate the three stable conformations—the anti-periplanar and the two synclinal conformations. These different stable conformations of butane are some sort of isomers. They are called *conformational isomers* or *conformers* for short.